SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (23 marks – approximately 41 minutes)

(a) Contrast a perfectly competitive market with a monopoly in terms of:

- **output decision**
- **pricing decision**
- **long-term profitability**

(6 marks)

(b) Consider a monopoly firm with demand and costs as shown below:

Demand curve: P = 25 - 2Q
Costs: MC = HK$19 and NO fixed cost

Required:

(i) Compute the price, marginal revenue and marginal cost in the following table.

Quantity (unit)	Price (HK$)	Marginal revenue (HK$)	Marginal cost (HK$)
1			
2			
3			
4			

(5 marks)

(ii) Given the output range stated in the table, state the profit-maximising output condition and determine the firm's profit-maximising output level accordingly.

(4 marks)

(iii) Based on the profit-maximising output level condition in part (b)(ii), determine the profit the firm would make.

(3 marks)

(iv) Define consumer surplus.

(2 marks)

(v) If the market is a perfectly competitive market (instead of a monopoly), compute the amount of consumer surplus that would be transferred to the monopoly.

(3 marks)

<u>Question 2</u> (20 marks – approximately 36 minutes)

(a) Contrast nominal gross domestic product ("GDP") with real GDP.

(4 marks)

(b) One of the most commonly used approaches in measuring an economy's GDP is the expenditure approach.

<u>Required:</u>

Analyse which of the expenditure components and how much (if at all) of the economy's GDP is affected in each of the following events.

(i) The government spends HK$50 million to import new police cars to replace old ones.

(4 marks)

(ii) A wrist-watch manufacturer focusing only on the domestic market produces HK$200 million worth of product, but consumers only purchase HK$140 million worth of them. (1 mark)

(4 marks)

(c) Analyse the impact of the following events on the aggregate demand ("AD") curve and/ or aggregate supply ("AS") curve and, hence, the price and output level in the short run.

(i) A fall in the size of labour.

(4 marks)

(ii) An appreciation of the domestic currency against currencies of the country's major trading partners.

(4 marks)

<u>Question 3</u> (5 marks – approximately 9 minutes)

(a) Define opportunity cost.

(2 marks)

(b) Given the information below, explain the opportunity cost (with calculations support) of doing a full-time top-up undergraduate degree at a college in your hometown while staying at your parent's place.

- **Tuition: HK$50,000**
- **Annual income from a 2-hours-per-week part-time job: HK$16,000**
- **Books: HK$1,000**

(3 marks)

<u>Question 4</u> **(20 marks – approximately 36 minutes)**

The following table shows a sample of the rates of return of two investment portfolios over a five-year period.

Year	Rate of return (Portfolio 1)	Rate of return (Portfolio 2)
2015	3%	21%
2016	35%	26%
2017	30%	18%
2018	27%	-5%
2019	-15%	45%

<u>Required:</u>

(a) (i) State which type of average return (arithmetic or geometric) should be used if you are trying to estimate the yearly return of 2020.

(1 mark)

(ii) Compute the arithmetic average rates of return of each investment portfolio.

(4 marks)

(b) (i) Compute the range of the rates of return of each investment portfolio.

(2 marks)

(ii) List any TWO advantages and any TWO disadvantages of the range as a measure of dispersion.

(4 marks)

(c) (i) Compute the standard deviations of the rates of return of Portfolios 1 and 2 for the period 2015 – 2019.

(2 marks)

(ii) Explain how standard deviation generally improves the usefulness of the range as a measure of dispersion.

(2 marks)

(d) (i) Compute the coefficient of variation ("CV") of the rates of return of Portfolios 1 and 2 for the period 2015 – 2019.

(2 marks)

(ii) Explain the general conditions under which CV will be a more appropriate measure of dispersion (of return) than standard deviation.

(3 marks)

<u>Question 5</u> **(12 marks – approximately 22 minutes)**

(a) **Explain the probability and non-probability sampling methods.**

(4 marks)

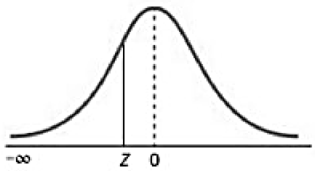
(b) **State and describe any THREE non-probability sampling methods.**

(9 marks)

* * * END OF EXAMINATION PAPER * * *

APPENDIX

The Cumulative Standardised Normal Distribution



Entry represents area under the cummulative standardised normal distribution from −∞ to Z

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−3.9	0.00005	0.00005	0.00004	0.00004	0.00004	0.00004	0.00004	0.00004	0.00003	0.00003
−3.8	0.00007	0.00007	0.00007	0.00006	0.00006	0.00006	0.00006	0.00005	0.00005	0.00005
−3.7	0.00011	0.00010	0.00010	0.00010	0.00009	0.00009	0.00008	0.00008	0.00008	0.00008
−3.6	0.00016	0.00015	0.00015	0.00014	0.00014	0.00013	0.00013	0.00012	0.00012	0.00011
−3.5	0.00023	0.00022	0.00022	0.00021	0.00020	0.00019	0.00019	0.00018	0.00017	0.00017
−3.4	0.00034	0.00032	0.00031	0.00030	0.00029	0.00028	0.00027	0.00026	0.00025	0.00024
−3.3	0.00048	0.00047	0.00045	0.00043	0.00042	0.00040	0.00039	0.00038	0.00036	0.00035
−3.2	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056	0.00054	0.00052	0.00050
−3.1	0.00097	0.00094	0.00090	0.00087	0.00084	0.00082	0.00079	0.00076	0.00074	0.00071
−3.0	0.00135	0.00131	0.00126	0.00122	0.00118	0.00114	0.00111	0.00107	0.00103	0.00100
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2388	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2482	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641